



Fig. 3: Track-via-missile guidance concept

ance to long-range missiles. A practical solution can be to use semi-active radar guidance or inertial guidance for mid-course correction and active guidance for terminal homing.

Raytheon's RIM-174 ERAM, also called standard missile-6 (SM-6), is a state-of-the-art surface-to-air missile in this category, having an operational range of 240km and using a combination of inertial guidance, semi-active radar guidance and active radar homing. It has been in service since 2013.

AIM-54 Phoenix air-to-air missile manufactured by Hughes Aircraft Co. and Raytheon Corp., with an operational range of 190km, also uses a combination of semi-active and active radar homing. It remained in service from 1974 to 2004.

MBDA Exocet anti-ship sea-skimming missile, with an operational range of 70km to 180km, also uses inertial and active radar homing guidance.

Track-via-missile radar guidance

TVM, also called re-transmission guidance, is a combination of semi-active radar homing and radio command guidance. In this case, the target is illuminated by an external radar, and the reflected radar energy is intercepted by a receiver onboard the missile, like it is done in semi-active radar homing.

However, the missile has no on-board computer to process these signals to generate track information. Signals are instead transmitted back to the radar-hosting launch platform over down data link for processing. The missile is then commanded over the up data link to adjust its flight trajectory to intercept the target.

Fig. 3 illustrates the concept of TVM guidance. MIM-104 Patriot missile system is an example of TVM guidance.

Unlike active radar guidance, which alerts the target aircraft about impending danger, in the case of TVM guidance, it may know that it is being illuminated but is never sure whether it is being engaged.

Modern phased-array radars, by virtue of their thin beams and low side-lobe levels, make detection by the aircraft even more difficult. TVM missiles can be made more accurate by using more sophisticated algorithms for calculating interception than would be possible in the limited processor possible to be accommodated in a missile.

In addition, operators have the option of adjusting the missile's flight path throughout engagement. Radar-hosting ground station may use radar energy reflected from the target directly and combine this information with the downloaded information from the missile to generate the interception course. This improves resistance to electronic countermeasures.

One disadvantage is vulnerability of the ground radar station to anti-radiation missiles as the radar has to remain active throughout engagement.

Possibility of data link jamming is another shortcoming.

Another potential disadvantage is that the missile will not be able to

continue engagement if the target aircraft could manage to put an obstacle such as a hillock between the radar beam and itself, or if it could manage to go out of the radar's tracking envelope.

Missile guidance and control

A missile's flight path is generally divided into three separate phases, namely, launching or initial phase, mid-course phase and terminal phase.

During the launch phase, flight controls are locked in a neutral position as the missile does not have aerodynamic stability during this phase. The guidance system takes charge immediately after the initial phase is completed and controls are unlocked. The initial phase lasts for a very short duration.

The main task of the mid-course guidance is to place the missile near the target, enabling the terminal phase guidance system to successfully take over. This phase is the longest in both distance travelled by the missile as well as time duration. During this phase, control may need to be exercised to ensure that the missile follows the desired course and also stays on course. In some cases, mid-course guidance may also additionally perform the task of terminal phase guidance.

Terminal phase is the most crucial phase of the missile's flight path, as it is this phase that leads to target-hit or target-miss. The terminal phase guidance system needs to have high accuracy and fast response to guidance signals. Towards the end of the missile's journey to intercept the target, when the missile is very close to target, it may not be left with sufficient energy to execute any high-g manoeuvre. Proportional guidance helps address this issue.

In the case of radar guidance, semi-active, active or TVM guidance, guidance signal is generated based on radar energy reflected off the target. Every missile-guidance system consists of an attitude control system and a flight path control system.